Failure Analysis Report

FAIL-0224 — Resource Exhaustion

1. Failure Summary

Failure ID	FAIL-0224
Failure Type	Resource Exhaustion
Severity	Minor
Component	UI Layer
Application	APP-006 — XRPlatform
Build Version	6.4.7
Timestamp	2024-02-02 18:44:07
Environment	ENV-005 — Infra Lab
OS Version	Ubuntu 22.04
Network	Ethernet
Resolution	Wont Fix

2. Test Case Reference

Test Case ID	TC-1145
Title	Test that NFC tap-to-pay correctly handles invalid input when Bluetooth interfer...
Execution ID	EXEC-01049
Test Run	RUN-006 — Regression Cycle 6 — Build 5.2.1
Requirement	REQ-0140 — Data Synchronization — Battery Optimization

3. Expected vs Observed Behaviour

Expected Behaviour	The system should complete biometric verification within the defined SLA.
Observed Behaviour	The system exhibited resource exhaustion when the battery level drops below 10%.

4. Error Details

Error Message:

```
OOMError: heap exhausted at 2045MB/2048MB during media_decode.
```

Stack Trace / Log Excerpt:

```
com.samsung.health.SearchIndexer.execute(line 222)
```

Log Reference	LOG-71510

5. Executed Steps at Time of Failure

1	Observe the status indicator for up to 11 seconds.	The device reconnects automatically within 10 seconds.

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2	Simulate network drop using the network emulator profile 'LTE-Thr...	A confirmation notification appears on the paired device.
3	Assert that database record for entity 'ENT-664' has status='FAIL...	No crash or ANR is observed.
4	Simulate Bluetooth interference using the network emulator profil...	No crash or ANR is observed.

6. Root Cause Analysis

Encryption key rotation not propagated to all microservice replicas before TTL expiry.

7. Linked Defect

Defect ID	BUG-0456
Title	Resource Exhaustion in UI Layer: OOMError: heap exhausted at 2045MB/2048MB durin...
Severity	Minor
Priority	High
Status	IN_PROGRESS
Root Cause	Encryption key rotation not propagated to all microservice replicas before TTL e...
Resolution	—
Assigned To	ENG-072

8. Historical Similar Failures

HIST-0132	FAIL-0133	Resource Exhaustion	3.4.2	2023-04-23	2	Resolved
HIST-0137	FAIL-0138	Resource Exhaustion	5.2.9	2022-08-28	12	Resolved
HIST-0223	FAIL-0224	Resource Exhaustion	6.1.9	2023-01-21	8	Resolved

9. Synthetic Data Notice

This document is entirely synthetic and intended for AI/RAG research only.

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